A REMARK ON OPENAI'S SMITH–TODA COMPLEX $V(4)$: FROM $p = 1009$ TO EVERY PRIME $p \geq 11$

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Abstract. OpenAI's preprint *A finite Smith–Toda complex $V(4)$ at the prime* 1009 chooses $p = 1009$ to make its finite inequalities generous. We check that every finite claim in that proof holds, as written with 1009 replaced by p, for all primes $p \geq 19$. At $p = 11, 13, 17$ three inequalities fail, but the statements they were used for can be verified directly by exact enumeration of May E_1 pages. **Conditional on the correctness of the original $p = 1009$ manuscript, which we did not re-verify**, $V(4)$ therefore exists at every prime $p \geq 11$. The argument is OpenAI's; we only re-select the prime and replace three inequalities by exact enumeration, and we list every changed line. Developed with AI assistance (Anthropic Claude, Opus 5.5) under [AUTHOR NAME]'s direction; independent adversarial reviews were run by separate AI sessions and are not human peer review.

1. Statement

Let [1] denote `preprints/A-finite-Smith-Toda-complex-V4-at-the-prime-1009-September-23-2026` of `https://github.com/openai/math` (commit `08fcd61`; line numbers refer to `build/sections/*.tex`) Its main theorem: at $p = 1009$ there is a finite p-local spectrum X with $BP_*X \cong BP_*/(p, v_1, v_2, v_3, v_4)$ as BP_*BP-comodules.

Theorem 1. *Assume that the proof in* [1] *is correct at $p = 1009$. Then for every prime $p \geq 11$ there is a finite p-local spectrum X with $BP_*X \cong BP_*/(p, v_1, v_2, v_3, v_4)$ as graded BP_*BP-comodules, with the canonical quotient coaction and generator in degree zero.*

2. Where p enters

The prime enters [1] only through finite statements about the May E_1 page $\Lambda(h_{i,j}) \otimes \mathbb{F}_p[b_{i,j}]$, with $h_{i,j}$ in bidegree $(1, p^j u_i)$, $b_{i,j}$ in $(2, p^{j+1} u_i)$ and $u_i = (p^i - 1)/(p - 1)$, together with explicit polynomial and divisibility inequalities used to prove those statements. We extracted every such inequality (33 in total; `ineq308.py` lists them with line numbers) and determined, for each, the primes at which it holds. All hold for $p \geq 19$. Exactly three fail somewhere in $\{11, 13, 17\}$:

line	inequality	holds for
`02` L109–113	$p^2 - \frac{27}{2}p - 9 > 0$	$p \geq 17$
`02` L154–155	no nonzero multiple of p in $[-14, 3]$	$p \geq 17$
`03` L161–163	$2p - 19 > p - 2$	$p \geq 19$

The first two are used only to prove Lemma `low:algebraic-vanishing`: certain cobar groups vanish because their May E_1 page is empty. The third is used only in the exhaustion proof of Table `may:n5-table`. At $p = 11, 13, 17$ we verify these conclusions directly. We enumerate every E_1 monomial in every relevant bidegree: $(qk + 1, u_n + k)$ for $n \leq 3$, and $(qk + 1 - \rho, U + k)$ for every profile $e \in \{0, 1, 2\}^{n+1}$ with $\rho \geq 2$, for all k up to the normalized cutoff of `low:normalized-cutoff`; also the bidegrees $(q + 1, u_4 + 1)$ and $(q + 1, u_5 + 1)$. The vanishing groups have no monomials. The table at $(q + 1, u_5 + 1)$ has exactly the five entries of [1], with

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exponents $p-2, p-3, p-3, p-3, p-4$ in place of $1007, \dots, 1005$, and d_1 is injective over $\mathbb{F}_p$ on the exterior-length-3 and length-5 spans. In the same runs we re-verify the other finite tables the proof uses at each p:

- the v_4-target lists, non-empty exactly at $k = 1, 2, p-1, p$, with the paper's monomials;
- the Frobenius quotient-group table;
- the small-stem lemma, with d_1-cohomology 0 at $(6, W+1)$;
- the page bound $qp + 1 - S = 2p - 6 > 0$, $N - K = p - 3 > 0$.

All other uses of 1009 in [1] are numerical instantiations of these symbolic inequalities (the README lists them line by line). The adversarial review read every section, including the appendix, and found no further dependence on the size of p, in particular no argument of the form "p is large, hence a group vanishes".

Remark 2 (Controls). At $p = 7$ the same enumeration finds non-empty product-obstruction pages, for example $y\,A^{16}$ in bidegree $(33, 119)$ for the profile $e = (0, 2, 0, 2)$, $k = 3$. So the mechanism fails at $p = 7$. This is consistent with Toda's requirement $p \geq 11$ for a unital product on $V(3)$ and with Nave's theorem that $V((p+1)/2) = V(4)$ does not exist at $p = 7$. At $p = 1009$ the enumeration reproduces the paper's tables verbatim (positive control).

Remark 3 (What is not claimed). We do not verify the homotopy-theoretic content of [1]: the adjusted-filtration injectivity (§3), the Frobenius left-hand side (§4), the extension of powers (§5) and the appendix pairing. Theorem 1 is exactly as reliable as that proof. Nothing is claimed for $p = 3, 5$.

3. Certificates

`may308.py` (direct recursive enumeration, exact integers, d_1 ranks over $\mathbb{F}_p$) and `may308_dp.py` (knapsack DP counts) form the primary computation. `independent/sweep.py` and `independent/d1check.py` were written separately; they use z3 exact enumeration with blocking clauses and an int64 DP modulo $2^{61} - 1$. All agree at $p = 7, 11, 13, 17$ and at $p = 1009$ (sampled). `may308.py` also passes at $p = 19, \dots, 47$.

References

[1] OpenAI, *A finite Smith–Toda complex $V(4)$ at the prime* 1009, OpenAI Math Release preprint, September 23, 2026, `https://github.com/openai/math/tree/main/preprints/A-finite-Smith-Toda-complex-V4-at-the-prime-1009-September-23-2026`.

[2] L. S. Nave, *The Smith–Toda complex $V((p+1)/2)$ does not exist*, Ann. of Math. 171 (2010), 491–509.

[3] H. Toda, *On spectra realizing exterior parts of the Steenrod algebra*, Topology 10 (1971), 53–65.